

Ayush Raj

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EDUCATION

Kalinga Institute of Industrial Technology (KIIT) <i>B.Tech, Computer Science & Engineering</i>	2022 – 2026 CGPA: 7.82
Jusco School Kadma, Jamshedpur <i>Class XII – PCM</i>	Percentage: 80.6%
Jusco School Kadma, Jamshedpur <i>Class X</i>	Percentage: 80.8%

INTERNSHIP EXPERIENCE

Tata Steel Ltd. (Shavak Nanavati Technical Institute) <i>Vocational Training Intern</i>	Apr 2025 – Jun 2025 Jamshedpur, India
- Analyzed industrial data handling protocols within Information Security & Data Protection division. - Designed and deployed the Smart Material Gate Pass System, automating entry/exit logging and reducing manual errors by 40% using Python, data validation, and logging pipelines. - Collaborated with cross-functional teams to enhance workplace security workflows. View Certificate	

RESEARCH & PUBLICATIONS

Generative Design for Personalized Fashion <i>Published – ICCCN 2025 (Springer)</i>	Bhubaneswar, India
- Co-authored a peer-reviewed paper on Generative AI in fashion using GANs and VAEs; proposed an AI-driven framework for personalized, sustainable clothing design. - Research explored data-driven design pipelines combining 3D garment modeling, user preference inputs, and generative models to automate custom clothing synthesis. Link to Paper	

TECHNICAL PROJECTS

SnapClass – AI Attendance System <i>Python, Streamlit, dlib, Resemblyzer, Supabase, Flask</i>	Live Demo
- Built a full-stack AI-powered attendance system using computer vision (dlib + SVM) and voice biometrics (Resemblyzer) to automate classroom roll calls. - Engineered a face recognition pipeline that detects multiple faces from a single classroom photo, computes 128D embeddings, and classifies students via an SVM classifier with a 0.6 distance threshold. - Implemented sequential voice identification using Librosa for audio segmentation and Resemblyzer for speaker embedding, enabling voice-based attendance from continuous recordings. - Deployed a decoupled architecture: Flask landing page on Vercel with SEO optimization and Streamlit app on Streamlit Cloud, both connected to a Supabase PostgreSQL database with QR-based enrollment.	
3D Portfolio Website <i>React, TypeScript, Three.js, React Three Fiber, GSAP, Rapier, Vite</i>	Live Link
- Built an immersive portfolio featuring a fully rigged 3D character rendered via Three.js + React Three Fiber with mouse-tracked head rotation, GLTF model loading, and ACESFilmic tone mapping. - Implemented a physics-based interactive tech stack using @react-three/rapier where skill-badge spheres respond to cursor collisions in real-time; section transitions powered by GSAP ScrollTrigger.	

TECHNICAL SKILLS

AI/ML & Computer Vision: Face Recognition (dlib), Voice Biometrics (Resemblyzer), scikit-learn (SVM), GANs, VAEs, TensorFlow, PyTorch, Librosa
Languages: C++, Python, JavaScript, SQL, Java, HTML, CSS
Frameworks & Tools: Streamlit, Flask, ReactJS, Next.js, Supabase, PostgreSQL, Tailwind, Git/GitHub, Vercel, VS Code
Core Concepts: Data Structures & Algorithms, OOPs, DBMS, REST APIs, System Design, Cloud Deployment

ACHIEVEMENTS & CERTIFICATIONS

- LeetCode:** Solved **150+ DSA problems** with focus on heaps, trees, dynamic programming, and linked lists.
- HackerRank SQL (Intermediate):** Certified for complex query writing including transactions, GROUP BY, and joins. [View Certificate](#)
- KIIT Merit Scholarship Holder:** Awarded for consistent academic performance across semesters.
- Springer ICCCN 2025 Publication:** Peer-reviewed research accepted and presented at an international conference.